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Harnessing Deep Neural Networks for Accurate Offline Signature Forgery Detection

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***Abstract*—Signature is considered to be one of the broadly practiced and important biometric features for the verification of an individual. It is used in various documents like pan card, passport, bank cheques, etc which can lead to numerous frauds like identity theft, money laundering and so on. To help solve or reduce this issue, an attempt is being made to develop an offline signature forgery detection system using an artificial neural architecture known as the Siamese neural network which consists of two identical Convolutional Neural Network (CNN) which are trained to learn feature representations of genuine and forged signatures. The two individual Siamese Models implemented in our project provide an accuracy of 98.27% and 98.35% respectively. Additionally, on occasions one of the models performs better while the other gets confused. To get optimum results from the former, a logical system was developed for handling the corner cases which resulted in an accuracy of 99.75%**

***Index Terms*—Artificial Intelligence, Machine Learning, Deep Learning, Optuna, Siamese Network, Convolutional Neural Network, Signature Forgery.**

I. INTRODUCTION

There are numerous biometric systems like facial recognition, fingerprint recognition and so on. Among these offline signatures are the most widely used feature in biometric systems [1] [2]. A signature is a distinctive way of writing one's initials or name to provide authorization in cheques, debit cards, property papers and so on. Forgery detection is the process of authenticating if a given signature matches with a person's original sign. In other words, signature verification is performed to avoid forgery while cashing cheques, transferring property or simply accepting couriers.

There are two types of signature verification systems: Online and Offline verification. Online verification makes use of features such as velocity, acceleration, movement of the pen, its pressure, and other dynamic features using a dynamic tablet. Offline verification on the other hand makes use of static features i.e scanned images of the signatures. Forgery detection using handwritten signatures is difficult as a forger can deliberately imitate a person's signature and hamper the authenticity of processes. Currently there is a person whose job is to match the signatures manually, this can cause two problems: the man in charge can be corrupt or the two signatures can look identical to a naked eye.

It is difficult for an individual to provide the exact same signature over multiple attempts and this discrepancy can be taken advantage of by a skilled forger [3]. The problem to be tackled here is to take into consideration the subtle changes in an individual's signature over multiple attempts and the fake signature a forger makes and finally to train a model which knows the difference between a genuine and a fake signature. To avoid this kind of forgery, a Siamese neural network [4] was constructed which consists of two identical convolutional neural networks (CNN) [5] [6] which are trained with the same weights to identify if a signature is real or fake. The dataset in use consists of multiple attempts made by a single person and also a bunch of forged signatures of the same person.

II. LITERATURE SURVEY

Poddar J,Parikh V, Bharti SK [7] proposed a technique for signature recognition using Convolutional Neural Networks (CNN) and Crest-Trough Method and signature forgery detection using SURF algorithm and Harris Corner Detection algorithm. They plan to increase the resultant system accuracy by trying new and better parameter coefficients that increase the deviation between real and forged signatures in the future.

H.-H.Kao and C.-Y.Wen [8] developed an offline signature verification system with an accuracy of 94.37% and 99.96% using Deep Convolutional Neural Networks (DCNN). However in order to optimize the system performance and

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provide a more convincing verification result they plan to collect more data and perform follow-up studies on them..

Navid S M A, Priya S H, Khandakar NH, Haque AB, Ferdous Z [9] constructed a model called VGG-19 which is constructed on pre-trained CNN. They made use of ICDAR as their dataset which yielded an accuracy of 100%, when CEDAR was used accuracy resulted to 88%, and 94.44% for Kaggle based dataset, to verify if a given signature is genuine or not. They faced computing resource limitations and were unable to work with larger datasets which they plan to overcome in the near future.

Hafemann L G, Sabourin R, Oliveira L S [10] approached the problem using feature learning perspective. Their hypothesis included learning features from the data instead of hand crafted ones which have no resemblance to the signature generation process. They intend to explore the importance of a good method for defining user-specific thresholds as their methods achieved low Equal Error rates. Also the actual False Rejection and False Acceptance rates are very imbalanced, and not stable across multiple users and datasets.

Xiao W, Ding Y [3] proposed a method using two-stage Siamese neural network model for accurate offline handwritten signature verification which adopts a two-stage Siamese neural network to verify original and enhanced handwritten signatures simultaneously, and also utilizes the Focal Loss to deal with the extreme imbalance between positive and negative offline signatures.

R. K. Bharathi B. H. Shekar [11] proposed a model which uses eigen-signature construction They have worked on converting scanned signature images to a shape form and eigen-signature construction is proposed for extracting the feature vector from a shape formed signature. The proposed system is conducted on a newly created Kannada offline signature database.

Boubacar Diallo, Thierry Urruty, Pascal Bourdon, Christine Fernandez-Maloigne [12] identified the problem of images present on online sharing platforms which have high probabilities of undergoing transformation like resizing, filtering which may impose many constraints on forgery detection algorithms. They proposed a camera identification model based on CNN.

Ane Bromley, Isabelle Guyon, Yann LeCun, Eduard Sickinger and Roopak Shah [13] proposed an algorithm for verifying signatures based on a Siamese neural network which consists of identical sub-networks joined at their outputs. They hope to have a field trial in the future which can be run in order to test this technology in the real world.

III. PROPOSED SYSTEM

A. Dataset Description

The ”Signature_Verification_Dataset” [14] which contains the signatures of Dutch Users for checking forgery is used to train the model was sourced from Kaggle [15], a popular data science community that provides a wide range of tools and resources for data analysis and machine learning. This dataset consists of signatures of Dutch users that were collected specifically for the purpose of detecting forgery. The signatures were obtained from the ICDAR 2011 Signature Dataset [16].

The offline dataset comprises PNG images that were scanned at 400 dpi and in RGB color, and was originally used in a signature verification competition organized by the German Research Center for Artificial Intelligence in collaboration with M Liwicki in 2011. The dataset is organized into two main directories: Train and Test. Each directory includes two subdirectories for each user, namely ”Genuine” (with the user number) and ”Forgery” (with the user number + ”_forg”). There are a total of 128 subdirectories (64 subdirectories containing genuine signature samples and 64 subdirectories containing forged signature samples of 64 users) and 42 subdirectories(21 subdirectories containing genuine signature samples and 21 subdirectories containing forged signature samples of 21 users) present in the Train and Test directories, respectively. Please note that each subdirectory may contain a variable number of signature samples. Additionally, the Train and Test directories do not overlap in terms of users, meaning that the users in the Train directory are distinct from those in the Test directory. No additional preprocessing or cleaning steps were performed on the dataset.

Additionally, the dataset includes two separate CSV files for the training and testing data, containing 23,205 and 5,747 rows respectively. These CSV files were developed by randomly selecting a combination of two images from either the genuine or forgery directories, belonging to either the same or different user. The CSV files are arranged systematically and consist of three fields: the path to the two images, and a label indicating whether the images are classified as genuine or forged.

B. Pre-Processing

In order to prepare input images for signature forgery detection using the Siamese network, it is necessary to perform pre-processing steps. These steps typically involve resizing, grayscaling, and binarization of the scanned images of signatures. The purpose of this pre-processing is to reduce the complexity of the input images and improve the computational efficiency of the Siamese network.

Image Resizing: To ensure consistency and improve the performance of the Siamese network, which includes a CNN, all input images are resized to a standard resolution of 128

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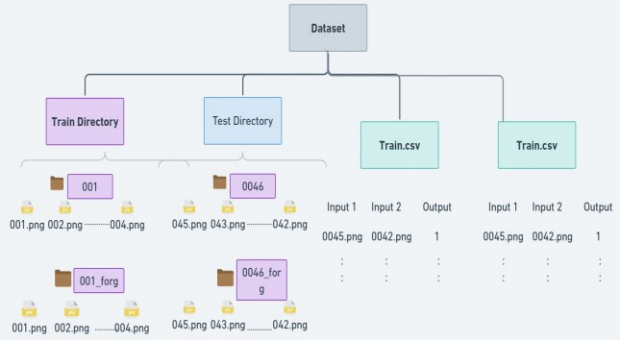


Fig. 1. Overview of Dataset Organization

x 128 before being passed through the network. Resizing the images improves computational efficiency, reduces noise, and prevents overfitting, while also making the model less sensitive to variations in the size and slant of the signature. This allows the model to more accurately detect forgeries.

Image GrayScaling: Image grayscaling is the process of converting a color image into a grayscale image, where each pixel in the grayscale image represents the luminance or brightness of the corresponding pixel in the original color image. This is achieved by taking the weighted average of the red, green, and blue color channels of an RGB image. Grayscaling is performed to simplify the image representation and reduce the computational cost required for processing. Additionally, it also contributes towards enhancing certain features of an image, such as edges and textures, making them easier to detect.

Image Binarization: Image binarization is a process that involves converting a grayscale or color image into a binary image by applying a threshold value. Pixels with values below the threshold are set to 0, while pixels with values above the threshold are set to 1. Although there are several methods to determine the appropriate threshold value for different applications, the OTSU thresholding method [17] is often used to determine it automatically. This pre-processing step is performed to standardize the pixel values, which can improve the performance of CNN models.

The train and test CSV files have been modified to replace the file path with the NumPy representation of the pre-processed images. Performing these steps has reduced the computational resources required to work with raw images and has also sped up the training and testing processes.

C. Model Implementation and Training

For the modeling aspect of this study, Keras (Version 2.6.0) [18] was utilized in conjunction with Tensorflow (Version 2.6.4) [19] as the backend. The neural network utilized for detecting forged signatures is constructed using a novel artificial neural architecture called the ”Siamese” neural network [13].

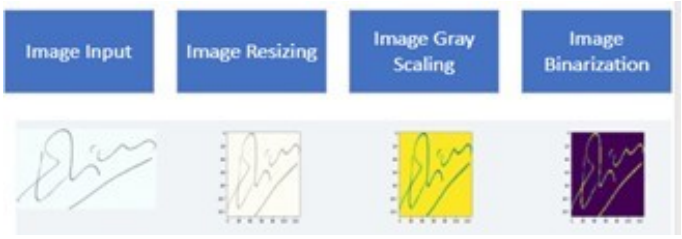


Fig. 2. Pre-processing workflow

The Siamese network consists of two identical convolutional neural networks (CNNs) that share the same weights and are trained to learn feature representations of genuine and forged signatures. The selection of a CNN-based approach for signature forgery detection is justified by its ability to learn discriminative features from the raw image data, and its proven success in a wide range of image recognition tasks. The output of the two CNNs (the pecuilar number of neurons present in each CNN layer is shown in Fig. 5 and Fig. 6 and the reason behind the selection of neurons is described latter part of this section) is then concatenated and passed through two dense layers that contain 64 neurons and 1 neuron respectively. These dense layers serve the purpose of introducing non-linearity to the feature representation learned by the CNNs. The first dense layer contains 64 neurons and uses the Rectified Linear Unit (ReLU) activation function. ReLU is a commonly used activation function in neural networks because it is computationally efficient and allows the model to learn complex non-linear relationships between features. The second dense layer contains a single neuron and uses the sigmoid activation function. This layer is responsible for computing the dissimilarity score between the two signature images. If the dissimilarity score is below a predefined threshold, the signature is classified as genuine, otherwise, it is classified as forged. A threshold value of 0.5 is chosen because it represents a natural decision boundary between the two classes. When the output of the Siamese network is greater than or equal to 0.5, it means that the dissimilarity between the two signatures is high, and the network classifies that the signature was forged. Fig. 3. illustrates the overall working of the Siamese network.

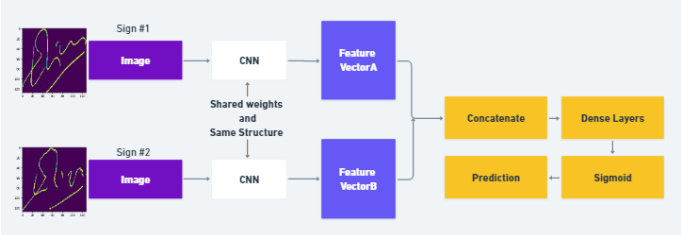


Fig. 3. Implementation of Siamese Network

Following our experimentation with various Convolutional Neural Networks, two architectures were identified that performed equally well on our validation data. As a result, two separate Siamese Network models were implemented. Both

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the architectures were taken into consideration because the predictions made by both Convolutional Neural Networks used in the two sub-networks of each Siamese model were dependent on the relative positions of the two images in certain input pairs, potentially affecting the accuracy of our model. Specifically, if the two images were presented in reverse order, the model might produce different outputs than when the images were presented in their original order. To ensure that our final model produced accurate results regardless of the input image positions, both the architectures were utilized and their outputs was combined using a custom function designed to yield the final prediction out of the four predictions obtained for each input pair. Further analysis of this custom function is discussed in the results section. The proposed system is demonstrated in Fig. 4., with the CNN architectures of the two distinct Siamese models shown in Fig. 5. and Fig. 6., respectively.

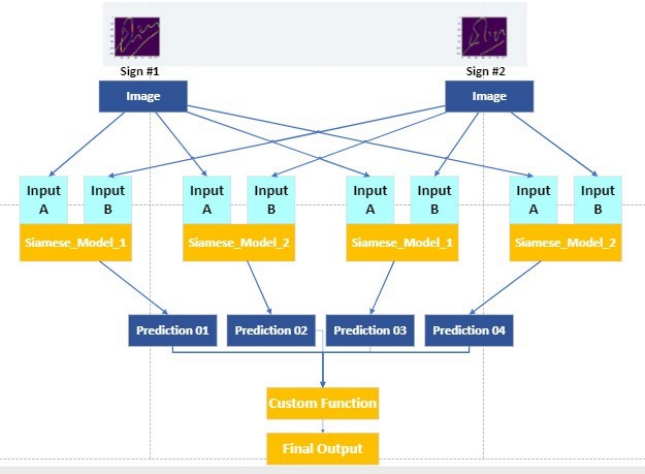


Fig. 4. Proposed System Architecture

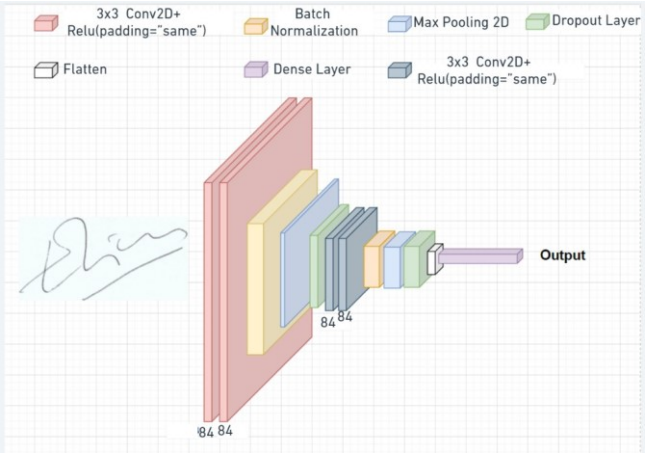


Fig. 5. CNN Architecture for Siamese_Model_1

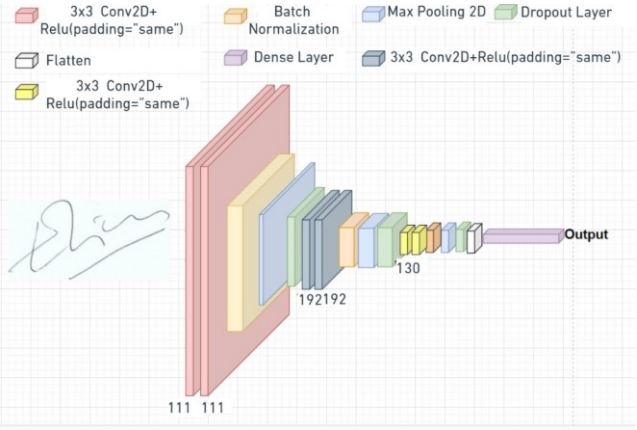


Fig. 6. CNN Architecture for Siamese_Model_2

The number of neurons in each convolutional layer depicted in Fig. 5. and Fig. 6. was determined through hyperparameter tuning, which was accomplished with Optuna, a hyperparameter optimization framework [20]. In the pursuit of maximizing validation accuracy, 20 trials for the hyperparameter search process were carried out. The study included the following parameters to obtain their optimal values:

- num_layers: An integer value between 1 and 3 indicating the number of layers in the model.
- num_units: A list of integer values between 64 and 256 indicating the number of units in each layer.
- num_epochs: An integer value between 1 and 15 indicating the number of epochs to train the model.
- dropout_rate: A float value between 0 and 1 indicating the dropout rate to be used in the layers.

In order to minimize the binary cross-entropy loss between predicted and target outputs, the model was fitted on the training dataset with a validation split of 0.2. The values of the parameters obtained from each of the 20 trials were then trained using the RMSprop optimizer (RMSProp is an unpublished adaptive learning rate optimizer proposed by Geoff Hinton.) [21] to maximize validation accuracy, with weights initialized and trained accordingly. The trained models were subsequently tested on the test data, and found that two convolutional layer architectures performed exceptionally well, not only on test data, but also on a few signature samples collected from study participants. As a result, two distinct Siamese models were implemented, each of which was trained for 5 epochs with a batch size of 128.

To make a prediction for a new input pair of signature images, first the images were preprocessed as described above and then pass them through both Siamese models. Each Siamese model produces two dissimilarity scores corresponding to the two possible input pair configurations. The four dissimilarity scores are then combined using a custom function (detailed description is present in ”Results and Analysis” Section) that takes into account the relative positions of the

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two images in the input pair. The output of the custom function represents the final binary classification prediction.

IV. RESULTS AND ANALYSIS

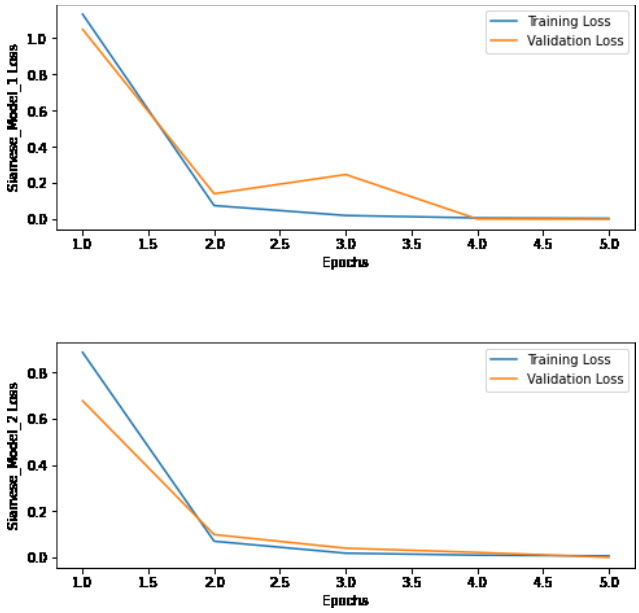


Fig. 7. Siamese _Model _1 and Siamese _Model _2 Loss

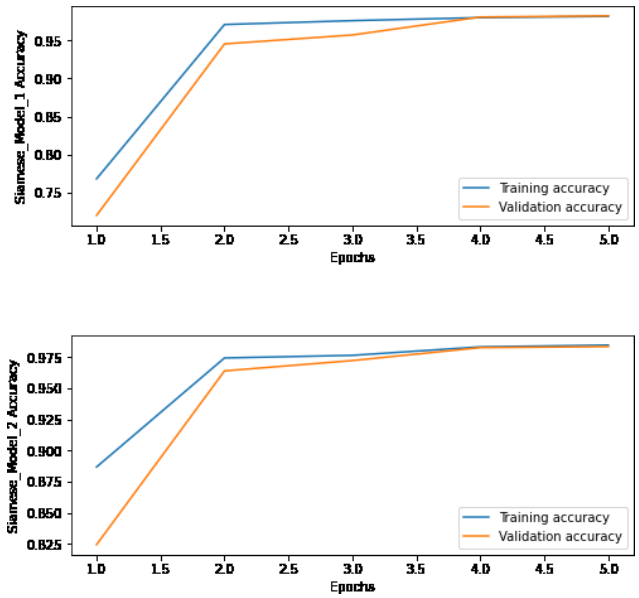


Fig. 8. Siamese _Model _1 and Siamese _Model _2 Accuracy

To assess the models' performance during training, the training and validation losses as well as the training and validation accuracies were tracked for each epoch. The Fig. 7 and Fig. 8 were generated and updated dynamically during

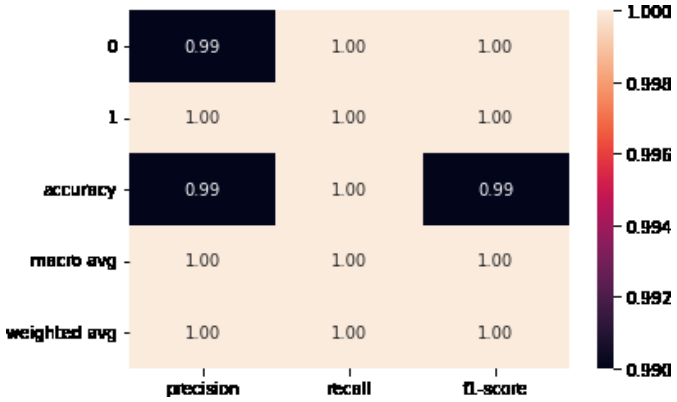


Fig. 9. Summary of Classification Report for the Proposed System

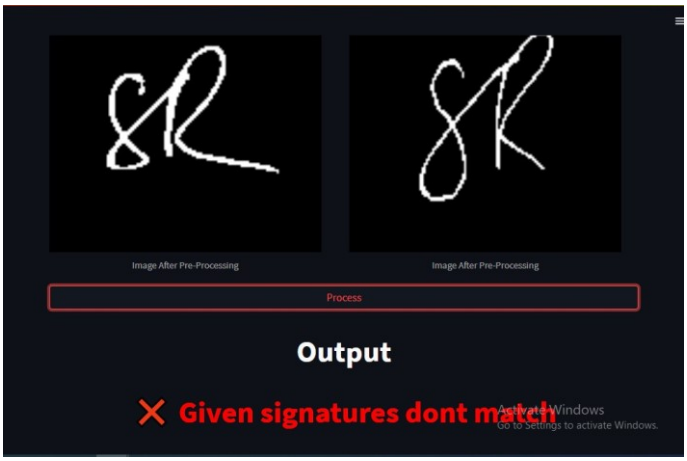


Fig. 10. Demonstration of the working of the model for Forged Signature Pair

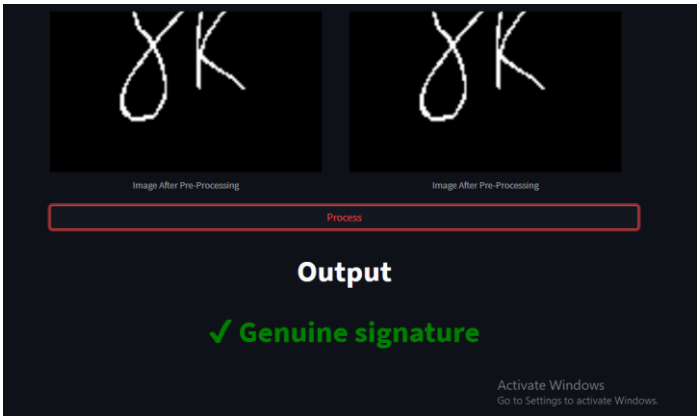


Fig. 11. Demonstration of the working of the model for Genuine Signature Pair

the model training by specifying the callback parameter while fitting the model. Tensorboard offers an interactive and user-friendly interface for monitoring and analyzing the model's performance, allowing for the visual representation of metrics like training and validation accuracy. By using Tensorboard's callback parameter, it was possible to generate these figures in real-time, to monitor the model's performance and adjust

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the hyperparameters as needed to achieve the best possible accuracy. Once each Siamese Model was trained independently, the resulting accuracy on the test files was 98.27% for Siamese_Model_1 and 98.35% for Siamese_Model_2. Based on the infrastructure described in Fig. 4. , when combining the predictions from two Siamese Models resulted in four predictions which gave rise to following cases which are handled by custom function to give the corresponding results in a following way:

Case 1:

When three out of four dissimilarity scores predicted are greater than 0.5, the final result is deemed as "forged.

Case 2:

When three out of four dissimilarity scores predicted are less than 0.5: The final result is deemed as "genuine"

Case 3:

If both the dissimilarity scores predicted by Siamese_Model_1 are below 0.5 and both the dissimilarity scores predicted by Siamese_Model_2 are above 0.5 or vice versa:

Example 1: If the predicted scores are [0.02, 0.48, 0.97, 0.98], where Siamese_Model_1 predicts 0.02 and 0.48, and Siamese_Model_2 predicts 0.97 and 0.98. To determine the final output, we need to consider the scores predicted by both models. The following steps were performed:

- The values below 0.5 are subtracted by 0 and the values above 0.5 are subtracted from 1.0. This ensures that all scores are within the expected range of 0.0 to 1.0. In the example considered, the modified scores become [0.02, 0.48, 0.03, 0.02].
- Next, we identify outliers that may indicate one model is more reliable than the other. In the modified scores calculated,it indicates that 0.48 is an outlier. This is because it aligns more with the predictions predicted by Siamese_Model_2 rather than Siamese_Model_1 itself, which shows that Siamese_Model_1 got confused. Since Siamese_Model_2 is reliable in this case and the majority of votes are for distance being greater, the result predicted by Siamese_Model_2, i.e., the output being "Forged," is returned.

Example 2: If the four predicted scores are like [0.39, 0.39, 0.61, 0.61] where 0.39 is predicted by Siamese_model_1 and 0.61 is predicted by Siamese_Model_2. In such a case, where both the predictions made by a particular model are the same and applying the steps specified in Example 1 fails. Therefore, more signature samples were requested. Such a case did not appear in test data so calculating accuracy did not have any problem. This case arised when testing with the signature samples provided by study participants.

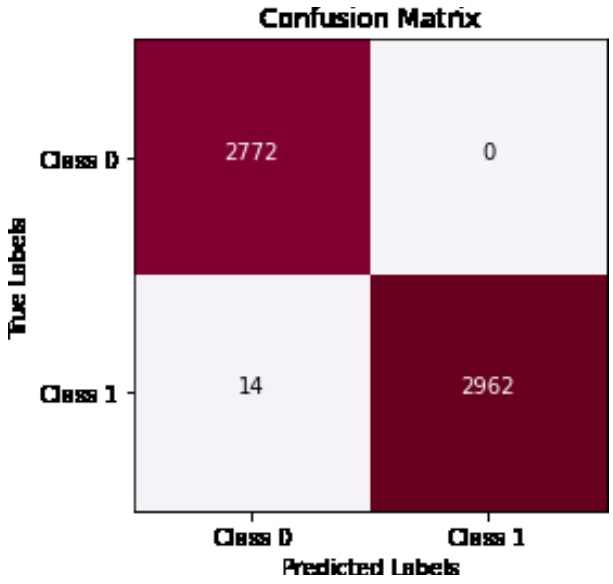


Fig. 12. Confusion Matrix Results for the Proposed System

Case 4:

If the four predicted scores are like [0.01, 0.97, 0.04, 0.98] where 0.01 and 0.97 are predicted by Siamese_model_1 and 0.04 and 0.98 are predicted by Siamese_Model_2: In such a scenario, It is possible that a Siamese model may have difficulty detecting forged signatures when both signatures provided are too similar. This is because the model relies on differences between the signatures to make its prediction, and when the signatures are too similar, those differences may be too subtle for the model to detect. In such cases, the model may struggle to accurately classify the input images and may make errors. However, if both Siamese models are predicting forgery for a certain relative position of input images, it is possible that the images are indeed forged. This is because each Siamese model has been trained to detect differences between signatures, and if both models are independently detecting the same differences and predicting forgery, it provides strong evidence that the signatures are not genuine.

After considering the final prediction using the custom function described above, the proposed system achieved an overall accuracy of 99.756%. The confusion matrix and classification report summary are presented in Fig. 9. and Fig. 10. for further details, respectively.

V. CONCLUSION AND FUTURE SCOPE

Our experimental results demonstrate that our Siamese network-based signature forgery detection system described in this study achieves a high level of accuracy (99.75%), indicating the effectiveness of our proposed approach in detecting signature forgeries. The novelty of our proposed model lies in the use of a Siamese network with two separate sub-networks.

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We utilized two distinct convolutional layer architectures for each sub-network to handle the issue of the relative positions of the two images in input pairs, which can affect the accuracy of the model. The final prediction is obtained by combining the outputs of the two sub-networks using a custom function that takes into account the relative positions of the two images. The neural network approach to automated signature forgery detection is a significant advantage, as it eliminates the need for manual inspection by experts, which can be time-consuming and costly. However, a disadvantage of this model is that the features it uses to produce the expected output are not known. Nevertheless, the model provides objective and consistent evaluation, unlike human experts who may be influenced by biases and contextual factors. Additionally, signature forgery detection models are useful for quickly and accurately verifying a large number of signatures, making them ideal for high-volume processing in banks, financial institutions, or legal proceedings.

The proposed system can be used to provide API services for signature authentication and verification. To authenticate a signature using an API, the user would need to provide two signature images intended for verification, along with any relevant information, such as the name of the signer or the date it was signed, and the API would then process the image and indicate whether the signature is genuine or forged.

To further enhance the reliability of the model, a hardware setup such as a tablet could be utilized to detect and verify signatures based on factors like pressure points and writing style. When a person signs their name on a tablet, the device captures various parameters like signature shape, movement, pressure points, speed, and acceleration of the stylus [22]. By incorporating these parameters and training the model with signature images, the credibility of the model's predictions can be significantly improved.

Our current signature classification model is designed to determine the authenticity of Dutch signatures and has been found to work well with English signatures as well, given the similarity between Dutch and English writing styles. However, for languages written from right to left, such as Arabic and Hebrew, the direction of strokes could be a crucial factor in determining the authenticity of the signature. Conversely, for languages like Chinese, Japanese, and Korean, the structure of characters and strokes could be more relevant in detecting forgery. To analyze these differences accurately, more data would be required, and the model could be further developed to cater to various languages.

VI. ACKNOWLEDGEMENTS

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